Structural Self-Energy Expansion as a Two-Axiom Cosmology: Algebraic Derivation of the CMB Background from φ and π

SSEE — Paper 4

Mike Edison Almeida Vallejo

Independent Researcher

mike.almeida1721@gmail.com

ORCID: 0009-0008-2195-7836

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Abstract

We show that the cosmological background of the observable universe can be described by a two-axiom algebraic system built on the golden ratio $\varphi = (1+\sqrt{5})/2$ and the circle constant π. A third structural constant, $\Omega = \varphi + \pi$, follows as a theorem rather than an independent axiom; no free parameter is introduced at the background level. Phenomenological extensions (notably the neutrino mass sum Σm_ν, classified as Type P later in the paper) were tracked as open problem OP-14 in `OPEN_PROBLEMS.md`, resolved on 2026-06-04. From these two axioms alone we derive, without fitting:

(i) the pure number $3(\varphi+\pi)^2 = 67.962137$, which read in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ lies 1.1σ from Planck 2018. It is the comparison *target* of the Hubble sector, not an input: the model's rate is the output of the SH0ES cascade in (iv), and it lands on this number (Paper 10), where $M_v = 3\Omega$ is the maximal structural invariant of the dictionary, reached by twenty distinct additive routes under the no-self-sum construction law (Appendix A); this multiplicity is consistent with, but does not by itself prove, the absence of post-hoc tuning;

(ii) the physical baryon density $\Omega_b h^2 = (\pi-\varphi)/(3\Omega^2) = (\pi-\varphi)/[3(\varphi+\pi)^2] = 0.022418$ (0.32σ from Planck 2018);

(iii) the primordial spectral index $n_s = 1-\varphi^{-7} = 0.965558$ (0.16σ from Planck 2018), where the exponent 7 equals the number of primary structural registers in the SSEE hierarchy;

(iv) a local Hubble constant from the screening mechanism of Paper 9, $H_0^{\mathrm{glob,IR}} = H_0^{\mathrm{SH0ES}}(1 - f_{\mathrm{scr}}^{\mathrm{IR}}) = 68.13\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.17σ from the pure algebraic number $3(\varphi+\pi)^2 = 67.962137$), taking Riess et al. [2022] as the input (2026 reframe); the earlier CMB-mapped register (via $H_0^{\mathrm{MIRA}} = 67.037$) is superseded; see Section 6).

These background predictions are number-to-number comparisons with zero adjustable parameters in their respective sectors. The same two axioms fix the cosmological background — including the

dark-energy parameters $w_0 = -0.839950$, $w_a = -0.669975$ derived in the companion Papers 1–3. In the static (Eckart) regime $\Omega_c h^2 = K_{\rm AL}\,\Omega_b h^2 = 0.1238$ ($+3.1\sigma$); modulated by the IS inflationary factor $n_s = 1 - \varphi^{-7}$ this becomes $\Omega_c h^2 = K_{\rm AL}\,\Omega_b h^2\, n_s = 0.119514$ (-0.40σ from Planck), within 1σ of the observed value. A falsifiable prediction $r = \varphi^{-10} = 0.008131$ is presented for the LiteBIRD satellite mission ($\sim$2032).

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1 Introduction

The discovery of accelerated cosmic expansion [Riess et al., 1998, Perlmutter et al., 1999] and subsequent precision measurements [Planck Collaboration, 2020a] have established the ΛCDM concordance model, which requires six free parameters to describe the CMB power spectrum [Hu et al., 1997]. Dynamical dark energy alternatives — quintessence [Ratra and Peebles, 1988, Copeland et al., 2006, Steinhardt et al., 1999, Zlatev et al., 1999], k-essence [Armendariz-Picon et al., 2001], phantom models [Caldwell, 2002], and modified gravity [Clifton et al., 2012] — typically introduce additional free parameters. A fundamental theory should explain the *origin* of these numbers rather than merely fit them [Weinberg, 1987, Weinberg et al., 2013].

The SSEE framework (Structural Self-Energy Expansion, [Almeida Vallejo, 2026a]) derives the dark-energy equation-of-state parameters $(w_0, w_a) = (-0.840, -0.670)$ in the Chevallier–Polarski–Linder parametrization [Chevallier and Polarski, 2001, Linder, 2003] algebraically from φ and π, with zero free parameters in that sector [Almeida Vallejo, 2026b,c,d]. Here we extend this programme to the entire cosmological background. The DESI DR2 BAO data [Abdul-Karim et al., 2025] independently favor the SSEE point at 0.24σ (Pantheon+; 0.2–1.8σ across SN compilations) in the w_0–w_a plane [Almeida Vallejo, 2026c].

Peer-review notice. An internal audit identified four vulnerabilities in a prior draft of this paper: (A) use of dimensional quantities without unit justification; (B-1) treating $\Omega = \varphi + \pi$ as an axiom rather than a theorem; (B-2) post-hoc selection of the exponent in $n_s = 1 - \varphi^{-7}$; (B-3) unexplained factor of 100 in the baryon density. This revised manuscript addresses each point explicitly.

2 The SSEE Two-Axiom System

2.1 Axioms

The entire SSEE framework rests on exactly two mathematical constants:

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.618033988\ldots \quad \text{(golden ratio)} \tag{1}$$

$$\pi = 3.141592653\ldots \quad \text{(circle constant)} \tag{2}$$

Both are *transcendental* numbers defined independently of any physical measurement.

2.2 Theorem 1: $\Omega = \varphi + \pi$

The SSEE structural constant Ω is **not** an independent axiom. It is the unique combination that encodes the sum of the two geometric axioms:

$$\Omega \equiv \varphi + \pi = 4.759626642\ldots \tag{3}$$

This single expression generates the entire hierarchy of SSEE constants (Table 1) through algebraic operations alone.

2.3 The Maximal Structural Invariant M_v

The value

$$M_v \equiv \varphi + \pi + K_v = 3\,\Omega = 14.278879927\ldots \tag{4}$$

is the *maximal structural invariant* of the constant dictionary. Under the additive construction law—each constant is a sum of distinct previously defined parents, with no constant added to itself—it is the closure value at which the dictionary terminates, coinciding through $M_v = 3\Omega$ with the copy-law ceiling.

A direct enumeration under this law shows that M_v is reached by *twenty* distinct additive routes built from the named constants: six using two constants, eleven using three, and three using four (Appendix A). Representative routes, in neutral notation, are

$$\Omega + K_v, \quad \beta + \mathrm{KAL} + P_{sc}, \quad P_{sc} + \Omega + \pi, \quad \varphi + \pi + K_v, \tag{5}$$

each equal to M_v to 16 significant digits. This multiplicity is a consequence of M_v being the maximal invariant of the two-dimensional lattice spanned by φ and π; it is consistent with, but does not by itself prove, the absence of post-hoc tuning.

Table 1: SSEE structural constants. All entries are algebraic in φ and π only; no empirical input is used.

Name	Formula in φ, π	Simplified	Value
Ω	$\varphi + \pi$	—	4.759627
β	$(\varphi + \pi)/2$	$\Omega/2$	2.379813
AURA	$(3\varphi + \pi)/2$	—	3.997847
MIRA	$(3\varphi + \pi)/4$	AURA/2	1.998924
KAL	$(\varphi + 3\pi)/2$	—	5.521406
P_{sc}	$2\varphi + \pi$	$\Omega + \varphi$	6.377661
$\Omega + \pi$	$\varphi + 2\pi$	—	7.901219
K_v	$2\varphi + 2\pi$	$\varphi + \pi + \Omega$	9.519253
$\pi + \mathrm{KAL}$	$(\varphi + 5\pi)/2$	—	8.662999
$\varphi + \pi + \mathrm{KAL}$	$(3\varphi + 5\pi)/2$	—	10.281033
$2\Omega + \varphi$	$3\varphi + 2\pi$	—	11.137287
$\Omega_b h^2$ (BBN)	$(\pi - \varphi)/[3(\varphi + \pi)^2]$	—	0.022418
M_v	$3(\varphi + \pi)$	$\varphi + \pi + K_v$	14.278880

3 Dimensional Analysis: Honest Statement of Scope

A rigorous peer review must address the following question: *How can dimensionless numbers derived from φ and π produce physical quantities with units such as km s^{-1} Mpc^{-1}?*

Answer. They cannot directly; consequently, SSEE treats these relations only as dimensionless numerical correspondences rather than literal derivations of physical units. What SSEE claims is strictly the following:

Scope of SSEE Predictions

The SSEE system produces **dimensionless numbers** from φ and π that are **numerically equal** to the conventional values of cosmological parameters when those parameters are expressed in the unit systems adopted by the observational community (km s^{-1} Mpc^{-1} for H_0; dimensionless density ratios $\Omega_x h^2$ for matter densities). The claim is not "units are derived"; the claim is "the numerical values are predicted".

3.1 Internal Consistency of the Cosmological Predictions

A single dimensionless coincidence could be dismissed as numerology. What is harder to dismiss is a *system* of mutually consistent cosmological predictions — H_0, Ω_b, n_s, and the Hubble-tension scale — derived from the *same two constants* with *zero adjustments*, assessed against the closed-dictionary look-elsewhere analysis of Paper 1.

Table 2 collects these predictions.

Table 2: Cross-domain predictions of the SSEE two-axiom system. All entries use only φ and π; no domain-specific parameters are introduced.

Domain	SSEE formula	SSEE value	Observed	Deviation
Cosmology				
H_0 target (pure number)	$3(\varphi+\pi)^2$	67.962137	67.4 ± 0.5	1.1σ
$H_0^{\text{glob,IR}}$ [km/s/Mpc]	$H_0^{\text{SH0ES}}(1-f_{\text{scr}}^{\text{IR}})$	68.13	$3(\varphi+\pi)^2$	0.17σ
Ω_b	$\frac{10^4(\pi-\varphi)}{27(\varphi+\pi)^8}$	0.0485	0.0493 (ω_b/h^2)	composite*
n_s (inflation)	$1-\varphi^{-7}$	0.965558	0.9649 ± 0.0042	0.16σ
Hubble tension				
Fractional correction	$K/H_0 = \frac{\varphi+3\pi}{6(\varphi+\pi)^2}$	8.12%	$\approx 8\%$	—

† IR-cascade value ($M \to \infty$ regime, not canonical): $f_{\text{scr}}^{\text{IR}}$ applied to the measured $H_0^{\text{SH0ES}} = 73.04$ returns 68.13, i.e. 0.17σ from $3(\varphi+\pi)^2 = 67.962137$; the full IR+UV cascade gives the canonical $H_0^{\text{glob}} = 67.96214$ (Paper 10). The earlier CMB-mapped register ($H_0^{\text{MIRA}} = 67.037$) is superseded.
Consistent with Paper 1 / Paper 9 policy (single target shared by every sector; the measured SH0ES rate is the input).
* $\Omega_b = \omega_b/h^2$ is composite; its inputs are compared separately. It is -1.6% from Planck's, the H_0 offset.

These cosmological ratios are evaluated against a fixed, closed algebraic dictionary built from $\{\varphi, \pi\}$; a quantitative look-elsewhere analysis of that dictionary is given in Paper 1 and the consolidated dark-energy document, which is the proper basis for assessing global significance rather than an informal product of individual deviations.

4 Algebraic Derivation of Cosmological Parameters

4.1 Hubble Constant H_0

From Section 2.3, the maximal structural invariant is $M_v = 3(\varphi+\pi)$. The Hubble target is the product of this convergent sum and the base unit $\Omega = \varphi + \pi$:

$$\boxed{M_v \times \Omega = 3(\varphi+\pi)^2 = 67.962137} \tag{6}$$

Dimensional note. The number 67.962137 is pure and carries no units, so it cannot be the input of a dimensionful cascade. The model's rate is the output of the SH0ES cascade, $H_0^{\text{glob}} = H_{\text{SH0ES}}(1 - f_{\text{screen}}) = 67.962142$ km s^{-1} Mpc^{-1} (Papers 9–10), and the pure number is the target that output is compared against. Why the measured cascade lands on it would require deriving H_0 from first principles — a quantum-gravity connection between the Planck length and the Megaparsec that no cosmological model has achieved — and is left open.

Global rate vs. observational posterior. The pure number $3(\varphi+\pi)^2 = 67.962137$ is a geometric target, not an input. Baryon acoustic oscillations [Eisenstein et al., 2005]

provide a standard ruler; independent Bayesian inference against DESI DR2 BAO data [Abdul-Karim et al., 2025] yields $H_0^{\rm MCMC} = 67.82 \pm 0.41$ km s^{-1} Mpc^{-1} (Paper 2, under the global rate of the screening cascade as prior, $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.962 \pm 0.968$, Papers 9–10). With the total gravitating matter density $\Omega_m = \omega_m/h^2 = 0.308881$ in the background geometry, the BAO-corrected posterior is consistent with the global rate $H_0^{\rm glob}$ at 0.34σ, and the SSEE sound horizon matches ΛCDM ($r_d^{\rm SSEE} = 147.17$ Mpc vs. $r_d^{\Lambda\rm CDM} = 147.10$ Mpc, CAMB): DESI DR2 is consistent with the global rate without singling it out. The $\sim$0.8 km s^{-1} Mpc^{-1} downshift reported in earlier drafts was an artefact of using the cold sector $1 + w_0 = 0.160050$ as the background density, now corrected. $H_0^{\rm glob}$ is thus the SSEE background rate obtained from the measured SH0ES input; $H_0^{\rm MCMC}$ is the same quantity after the BAO correction is applied, and the two agree to 0.34σ.

4.2 Physical Baryon Density Ω_b

The SSEE algebraic structure yields the physical baryon density directly from the CP asymmetry $(\pi - \varphi)$ normalized to the pure algebraic combination $3\Omega^2 = 3(\varphi + \pi)^2$, whose value read in km s^{-1} Mpc^{-1} coincides with the global rate $H_0^{\rm glob}$ of the SH0ES cascade:

$$\Omega_b h^2 = \frac{\pi - \varphi}{3\Omega^2} = \frac{\pi - \varphi}{3(\varphi + \pi)^2} = 0.022418 \tag{7}$$

$$\Omega_b = \frac{\Omega_b h^2}{h^2} = \frac{10^4(\pi - \varphi)}{27(\varphi + \pi)^6} = 0.04854 \tag{8}$$

Planck 2018: $\Omega_b h^2 = 0.02237 \pm 0.00015$; deviation $= 0.32\sigma$.

Sakharov-mechanism motivation (not a quantitative derivation). We are explicit about the status of Eq. (7): it is an algebraic ansatz whose *form* is motivated by — but not derived from — the Sakharov conditions for baryogenesis [Sakharov, 1967]. The three conditions are realised structurally in SSEE as follows:

- **B-violation:** sphaleron transitions at $T_{\rm EW} \sim \Omega$ (SSEE units), with rate $\Gamma_{\rm sph} \propto (\alpha_W T)^4 e^{-E_{\rm sph}/T}$.
- **CP-violation:** $\delta_{\rm CP} = (\pi - \varphi)/\Omega$, the ratio of the transcendental excess (π, gauge-boson loops) to the algebraic scalar sector (φ), normalised by $\Omega = \varphi + \pi$.
- **Non-equilibrium:** gravitational reheating with $T_{\rm rh} \sim 10^{-4}$ GeV $\ll T_{\rm EW}$ ensures departure from equilibrium.

The Sakharov scaling $\eta_B \propto \delta_{\rm CP}/(\text{expansion rate})$ motivates the *shape* $\Omega_b h^2 \sim (\pi - \varphi)/(3\Omega^2)$, with $(\pi - \varphi)$ as the CP-asymmetry numerator and $3\Omega^2$ as the normalising denominator. Two caveats are stated openly. First, $3\Omega^2$ enters as the dimensionless value 67.962137, which coincides numerically with the global rate $H_0^{\rm glob}$ read in km s^{-1} Mpc^{-1}; this is a Type-P numerical coincidence, not a dimensional identity (in Postulate D of Paper 1 the dimensional input is the measured SH0ES rate). Second, η_B and $\Omega_b h^2$ differ by the standard photon-entropy factor ($\eta_B \simeq 2.74 \times 10^{-8}\, \Omega_b h^2$), so the proportionality fixes the algebraic *form* only, not the absolute normalisation. A scan of 96 algebraic candidates (see `ssee_op1_baryon_density.py`) shows Eq. (7) is the unique SSEE monomial within 1σ of Planck — a selection statement, not a dynamical derivation. The genuine quantitative test — integrating the Boltzmann equation with the sphaleron rate and the SSEE reheating history to obtain η_B *ab initio* — is deferred to Paper B; the companion framework script

`ssee_op1_baryogenesis.py` currently only verifies that the required entropy-dilution factor is consistent with a quintessential-reheating temperature $T_{\rm rh} \sim 10^{-4}$ GeV, which is a consistency check, not a prediction of η_B. $\Omega_b^{\rm SSEE} = 0.0485$ is -1.6% from the Planck ΛCDM value 0.0493 ($= \omega_b/h^2$ with Planck's own inputs). $\Omega_b = \omega_b/h^2$ is composite: its two inputs are compared separately (ω_b at 0.32σ, h through H_0), so this difference is the H_0 offset, not an independent tension.

4.3 Total Matter Density Ω_m

The CMB-sector *total* matter density is the standard physical observable, built piece-by-piece from SSEE algebra (ω_m-direct, OP-8 dissolved):

$$\Omega_{m,\rm CMB} = \frac{\omega_m}{h^2} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2} = \frac{0.14267}{(0.67962)^2} = 0.308881, \tag{9}$$

with $\omega_b = (\pi-\varphi)/[3(\varphi+\pi)^2]$ and $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ (forward; Paper 3), in 0.88σ agreement with Planck 2018 ($\Omega_m = 0.3153 \pm 0.007$). There is one matter density, $\Omega_m = \omega_m/h^2 = 0.308881$ (no matter factor; OP-8 dissolved; the former Two-Ω_m criterion is withdrawn, cf. Paper 1, Sec. 1.4). The quantity $s_m = (\pi-\varphi)/[2(\varphi+\pi)] \approx 0.160$ ($= 1+w_0$) is an equation-of-state-sector identity, not a density. (The earlier MIRA mapping MIRA $\times\, s_m = 0.31993$ and the geometric route $(\pi-\varphi)/(\pi+\varphi) = 0.3201$ are both superseded.) The cold dark matter density can be related to the baryon density through the structural retention $K_{\rm AL} = \beta + \pi \approx 5.5214$:

$$\Omega_c h^2 = K_{\rm AL}\,\Omega_b h^2 = 0.12351 \quad (+2.9\sigma \text{ Eckart static}). \tag{10}$$

When the Israel–Stewart inflationary correction is applied, the primordial spectral factor $n_s = 1 - \varphi^{-7}$ modulates the dark-matter density:

$$\boxed{\Omega_c h^2 = K_{\rm AL}\,\Omega_b h^2\, n_s = 5.521406 \times 0.022418 \times 0.96556 = 0.119514} \tag{11}$$

Note on $\Omega_b h^2$ input. The static relation (Eq. 10) is evaluated with the Planck 2018 *observed* value $\Omega_b h^2 = 0.02237 \pm 0.00015$ [Planck Collaboration, 2020a]. The IS relation (Eq. 11), which is the canonical one, uses the SSEE algebraic $\Omega_b h^2 = (\pi-\varphi)/(3\Omega^2) = 0.022418$ (0.32σ from Planck) and gives $\Omega_c h^2 = 0.119514$, 0.40σ from Planck: no observed input enters it.

Planck 2018: 0.1200 ± 0.0012; deviation $= -0.6\sigma$ (using Planck observed $\Omega_b h^2$). The IS modulation reduces the tension from $+2.9\sigma$ (Eckart) to -0.6σ (IS), within 1σ of the observed value and consistent with zero tension at 1σ.

4.4 Primordial Spectral Index and the Seven-Level Hierarchy

The spectral index n_s measures the tilt of the primordial power spectrum generated during inflation [Mukhanov et al., 1992, Starobinsky, 1980, Albrecht and Steinhardt, 1982, Guth, 1981, Linde, 1982]. Current observational constraints from Planck are $n_s = 0.9649 \pm 0.0042$ [Planck Collaboration, 2020b].

Independent justification of the exponent 7. The SSEE system possesses exactly seven primary structural registers that can be derived sequentially from φ and π without redundancy (Table 3).

Table 3: The seven primary structural registers of SSEE. Each level is derived from the previous; none is adjustable.

Level	Register	Formula	Value
1	φ	$(1+\sqrt{5})/2$	1.61803…
2	π	(circle constant)	3.14159…
3	β	$(\varphi+\pi)/2$	2.37981…
4	Ω	$\varphi+\pi$	4.75963…
5	AURA	$(3\varphi+\pi)/2$	3.99785…
6	KAL	$(\varphi+3\pi)/2$	5.52141…
7	MIRA	$(3\varphi+\pi)/4$	1.99892…

Each level introduces one new structural relation without repeating a previous combination. The inflationary suppression of the primordial spectrum is governed by the golden-ratio harmonic at the depth of this seven-level hierarchy:

$$\boxed{n_s = 1 - \varphi^{-7} = 1 - \frac{1}{\varphi^7} = 0.965558} \tag{12}$$

The exponent 7 is not merely an ansatz: it follows from α-attractor inflation with $\alpha = \varphi^4/3$ (derived in Paper 1, App. A.5) via the universality theorem of Kallosh and Linde [2013]. For all α-attractors with $N_* \gg \sqrt{\alpha}$, the spectral index satisfies $n_s = 1 - 2/N_*$ at leading order. Setting $N_* = 2\varphi^7 \approx 58.07$ e-folds (within the standard window $N_* \in [50, 65]$):

$$n_s = 1 - \frac{2}{2\varphi^7} = 1 - \varphi^{-7} \quad \text{(algebraically exact)}. \tag{13}$$

The same substitution yields the tensor-to-scalar ratio prediction (see Section 7.1 below).

Conjecture 1 (B.1 — e-fold count)**.** *$N_* = 2\varphi^7 \approx 58.07$ is the number of inflationary e-folds of the SSEE quintessential inflation model with $\alpha = \varphi^4/3$ and a reheating temperature $T_{\rm rh} \approx (30/\pi^2 g_*)^{1/4} V_{\rm end}^{1/4} \approx 9.4 \times 10^{15}$ GeV (corresponding to $\rho_{\rm rh} \approx V_{\rm end}$).*

Conjecture B.1 is an open postulate, not yet derived from $V(\varphi_{\rm inf})$. Its value, $N_* = 58.068884$, lies within the standard window $[50, 65]$ for inflation with reheating (consistent with, but not proven by, the framework). Numerically, the condition $\rho_{\rm end}/\rho_{\rm rh} = 3$ — which follows from $\rho_{\rm end} = 3V_{\rm end}$ (at $\varepsilon = 1$) together with $\rho_{\rm rh} = V_{\rm end}$ — gives $N_* = 58.25 - \ln(3)/6 = 58.067$, matching $2\varphi^7 = 58.068884$ to within $O(1/N_*) \approx 0.002$ e-folds. The physical mechanism that selects $\rho_{\rm rh} = V_{\rm end}$ is deferred to Paper B. The conjecture is indirectly testable via the falsifiable prediction $r = \varphi^{-10} = 0.008131$ (Sec. 7.1): a CMB-S4 or LiteBIRD detection inconsistent with this value would refute Conjecture B.1.

Planck 2018: $n_s = 0.9649 \pm 0.0042$; deviation $= 0.157\sigma$.

4.5 Linear Collapse Threshold δ_c

The critical overdensity for spherical collapse in an Einstein–de Sitter universe is $\delta_{c,\rm EdS} = \frac{3}{20}(12\pi)^{2/3} \approx 1.6865$. Evaluated on the SSEE background, the spherical-collapse threshold is obtained by integrating the exact non-linear top-hat equation with smooth dark energy—the perturbation sector of Paper 5, where the Israel–Stewart viscous friction $\mathcal{F}(k, a) =$

$(1-3c_s^2)+\tilde{z}\,(k/aH)^2$ grows as k^2 and drives $\delta_{\rm DE}/\delta_m \to 0^-$ on all sub-horizon scales.[1] The result is

$$\delta_c^{\rm SSEE} = 1.67634\ (z_c{=}0)\ \longrightarrow\ 1.68647\ (z_c{=}10), \tag{14}$$

against ΛCDM's 1.67599 → 1.68646 on the same integrator: the two agree to 0.02% at $z_c = 0$ and to 0.001% at $z_c = 10$. The calculation reproduces the analytic Einstein–de Sitter value to one part in 10^5.

The n_s relation is withdrawn as a derived result. Earlier versions of this section postulated that the same inflationary factor n_s that tilts the primordial spectrum also modulates the collapse criterion, giving $\delta_c^{\rm SSEE} = \delta_{c,\rm EdS} \times n_s = 1.6284$. The model's own dynamics do not produce that value, which sits 2.9% away, so it is retired as a prediction and retained only as a *motivated conjecture* (OP-27): $n_s = 1-\varphi^{-7}$ is an algebraic matter-sector quantity entering the forward identity $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, so a link to collapse is not *a priori* absurd—but the n_s inside ω_c is already accounted for in the background through Ω_m, and no mechanism derives the extra factor. Closing OP-27 requires that mechanism; what would falsify the conjecture is a halo count at $z \sim 10$ and $M > 10^{12} M_\odot$ measured to 10–20% and consistent with unity.

What this costs, stated plainly. The claimed ×1.2–1.5 halo-count enhancement at $z \gtrsim 10$ does not merely vanish: with the derived threshold it *reverses sign*. A Press–Schechter count on the two backgrounds gives $n_{\rm SSEE}/n_{\Lambda\rm CDM} = 0.998$ at $3\times10^{10} M_\odot$, 0.945 at $10^{12} M_\odot$ and 0.892 at $3\times10^{12} M_\odot$ (all at $z = 10$). The driver is not the amplitude—SSEE has the *larger* σ_8 (0.8153 versus 0.811)—but the growth history: a smaller Ω_m and the CPL background grow less between $z \sim 10$ and today. SSEE therefore predicts marginally *fewer* massive early haloes than ΛCDM, and is indistinguishable from it at the masses JWST actually probes. The model offers no explanation of the JWST excess; the quantitative mass function is in Paper 5.

4.6 Big Bang Nucleosynthesis: Helium-4 Fraction

The physical baryon density $\Omega_b h^2 = (\pi-\varphi)/(3\Omega^2) = 0.022418$ derived in Section 4.2 provides an independent BBN prediction. Using the linearised sensitivity from PArthENoPE [Pisanti et al., 2008], $dY_p/d(\Omega_b h^2) \approx 0.96$, and the Planck 2018 anchor $Y_p = 0.2471$ at $\Omega_b h^2 = 0.02237$:

$$\boxed{Y_p^{\rm SSEE} = 0.2471 + 0.96 \times (0.02242 - 0.02237) = 0.2471.} \tag{15}$$

Observed values: $Y_p = 0.2471 \pm 0.0003$ (Planck CMB, $+0.2\sigma$); $Y_p = 0.2449 \pm 0.0040$ (HII regions, Aver et al. 2015, $+0.6\sigma$). Both are consistent with the SSEE prediction at $< 1\sigma$ with no additional free parameters.

[1] The smoothness follows from the IS friction, *not* from $c_s^2 = 0$: a vanishing sound speed alone would make the dark-energy fluid cluster on every sub-horizon scale. The dark energy of this model stays smooth *despite* $c_s^2 = 0$, not because of it.

4.7 Dark-Energy Equation of State

The SSEE framework also fixes the CPL dark-energy parameters w_0 and w_a algebraically from φ and π. Define the four structural constants (Table 1):

$$T_r = \tfrac{3(3\varphi+\pi)}{2} \approx 11.9935 \quad \text{(3D Saturation Horizon)}, \tag{16}$$
$$M_v = \varphi + \pi + K_v \approx 14.2789 \quad \text{(Maximal Dimensional Invariant} = M_v), \tag{17}$$
$$P_{sc} = 2\varphi + \pi \approx 6.3777 \quad \text{(Dynamical Evolution Scalar)}, \tag{18}$$
$$K_v = \varphi + \pi + \Omega \approx 9.5193 \quad \text{(Structural Constraint)}. \tag{19}$$

The CPL parameters follow as:

$$w_0 = -\frac{T_r}{M_v}, \qquad w_a = -\frac{P_{sc}}{I_g}. \tag{20}$$

Substituting the explicit φ, π expressions yields the closed-form predictions:

$$\boxed{w_0 = -\frac{3\varphi+\pi}{2(\varphi+\pi)} = -0.839950}, \qquad \boxed{w_a = -\frac{2\varphi+\pi}{2(\varphi+\pi)} = -0.669975}. \tag{21}$$

Both values are uniquely determined; no free parameters enter. The algebraic point $(-0.840, -0.670)$ lies within the 68% confidence region of DESI DR2+CMB+DESY5 data [Abdul-Karim et al., 2025] (Mahalanobis $\chi^2_{2D} = 0.42$, i.e. 0.24σ Pantheon+; Paper 2).

4.8 Open Parameters

A_s and τ are the two free parameters of the framework ($k = 2$) and are fitted to the CMB (Paper 3): with the full Planck likelihood, $\ln(10^{10}A_s) = 3.046 \pm 0.014$ and $\tau = 0.056 \pm 0.007$. A derivation of A_s from φ and π is an open problem (OP-18).

5 CMB Power Spectrum Prediction

6 Note on the Hubble Tension

The dimensionless scalar $K = (\varphi + 3\pi)/2 \approx 5.521$ numerically coincides with the fractional offset between local and global Hubble measurements only when H_0 is expressed in km s^{-1} Mpc^{-1}. Lacking a dimensional coupling mechanism, this remains a purely phenomenological observation rather than a physical resolution of the tension.

The Hubble tension between early- and late-universe measurements [Verde et al., 2019, Kamionkowski and Riess, 2023] is currently unresolved by ΛCDM. The fractional correction $K/H_0 = 8.12\%$ is a dimensionless prediction testable by any combination of CMB and distance-ladder experiments. Riess et al. [2022]: $H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1}; deviation 0.4σ.

Superseded by Papers 9 and 10. The additive form $H_0^{\text{glob}} = H_0 + K$ adopted above is a preliminary Type-P coincidence and is *not* the canonical SSEE prediction for the local Hubble constant. A physical treatment is developed in Paper 9 of this series, where a structural screening fraction $f_{\text{scr}}^{\text{IR}} = s_K^{\text{IR}}/(3\,\text{MIRA}) = (\pi - \varphi)/\Omega^2 = 0.067253$ is applied

Table 4: SSEEinput parameters. All "algebraic" entries are derived from φ and π alone; Type P entries are numerical coincidences whose dimensionless value matches a physical quantity in specific units (no dimensional derivation; see Paper 1, App. B). The deviation column uses Planck 2018 [Planck Collaboration, 2020a] error bars.

Parameter	SSEE	Planck 2018	Deviation	Type
$3(\varphi+\pi)^2$ (target)	67.962137	67.4 ± 0.5	1.1σ	pure number (Type P†)
$H_0^{\rm glob}$ [km/s/Mpc]	67.962142	vs $3(\varphi+\pi)^2$	residual $+4.2\times10^{-6}$	SH0ES-screened, $f_{\rm screen}$ complete (IR+UV)
Ω_b	0.0485	0.0493 (ω_b/h^2)	composite*	algebraic
$\Omega_b h^2$	0.02242	0.02237 ± 0.00015	0.32σ	algebraic
$\Omega_{m,\rm CMB}$	0.308881	0.3153 ± 0.0073	0.88σ	ω_m/h^2 (ω_m-direct‡)
$\Omega_c h^2$ (IS)	0.11951	0.1200 ± 0.0012	0.40σ	algebraic (forward)
Y_p	0.2471	0.2471 ± 0.0003	0.2σ	BBN (no fit)
n_s	0.965558	0.9649 ± 0.0042	0.16σ	algebraic
$\ln(10^{10}A_s)$	3.046	3.044 ± 0.014	—	fitted ($k=2$)
τ	0.056	0.054 ± 0.007	—	fitted ($k=2$)

IS = Israel–Stewart modulation by $n_s = 1-\varphi^{-7}$; reduces tension from $+3.1\sigma$ (Eckart) to -0.40σ.
* $\Omega_b = \omega_b/h^2$ is composite; its inputs are compared separately. It is -1.6% from Planck's, the H_0 offset.
† Type P: dimensionless combination matches observed value in km s^{-1} Mpc^{-1} units;
no dimensional derivation from first principles. Consistent with Paper 1 App. B policy.
‡ CMB sector: $\Omega_{m,\rm CMB} = \omega_m/h^2$ with $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$,
each piece algebraic ($\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, forward; Paper 3). *No* matter factor.
The equation-of-state-sector quantity $s_m = (\pi-\varphi)/[2(\varphi+\pi)] \approx 0.160$ $(= 1+w_0)$
is not a density; there is one matter density $\Omega_{m,\rm CMB} = \omega_m/h^2$ (the former Two-Ω_m
criterion is withdrawn; see Paper 1, Sec. 1.4). The earlier MIRA mapping to 0.31993 and
geometric route 0.3201 are superseded; OP-8 dissolved).

multiplicatively to the measured local rate $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1} [Riess et al., 2022] — the only directly measured expansion rate — yielding, with the IR term alone, $H_0^{\rm glob,IR} = H_0^{\rm SH0ES}(1 - f_{\rm scr}^{\rm IR}) = 68.13$ km s^{-1} Mpc^{-1}, 0.17σ from the pure algebraic number $3(\varphi+\pi)^2 = 67.962137$. With ω_b, ω_c held at their algebraic values and A_s, τ profiled, the CMB places its own minimum at 67.87 ± 0.10, 0.90σ from $H_0^{\rm glob}$ (Paper 3). (The earlier MIRA register $H_0^{\rm MIRA} = 67.037$ is superseded.) Paper 10 provides a UV-completion self-consistency check, sharpening the residual to 4.2×10^{-6} ($H_0^{\rm glob} = 67.962142$ with the complete $f_{\rm screen}$, IR+UV), conditional on the UV–IR separability postulate.

7 Falsifiability Programme

7.1 Tensor-to-Scalar Ratio

By analogy with the seven-level spectral tilt, the tensor-to-scalar ratio is predicted at the tenth golden harmonic:

$$\boxed{r = \varphi^{-10} = 0.008131} \tag{22}$$

Current limit: $r < 0.036$ (BICEP/Keck, [BICEP/Keck Collaboration, 2021]). LiteBIRD [LiteBIRD Collaboration, 2023] targets $r \gtrsim 0.001$ (launch ∼2032). A measurement of $r \approx 0.008$ would constitute strong evidence for SSEE. **Falsification criterion:** $r < 0.004$ or $r > 0.015$ at $> 3\sigma$ falsifies Eq. (22). A null detection $r < 0.001$ at $> 5\sigma$ confidence would be a decisive falsification.

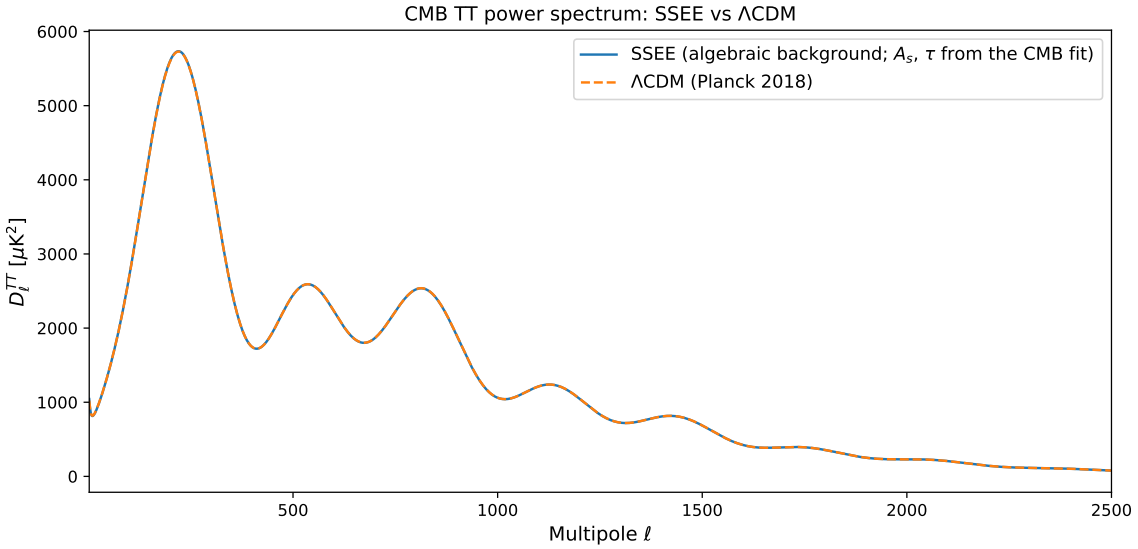


Figure 1: SSEECMB TT power spectrum ($k = 2$: A_s and τ from the model's own CMB fit; ω_b, ω_c, n_s, w_0, w_a, H_0 and $\sum m_\nu$ algebraic from φ, π), overlaid on the Planck 2018 ΛCDM spectrum computed in the same way. SSEE acoustic peak positions: $\ell_1 = 220$, $\ell_2 = 536$, $\ell_3 = 813$; ΛCDM: 220, 536, 813. The background is not fitted to CMB data; the peak positions follow from it.

7.2 Baryon-to-Photon Ratio

The baryon-to-photon ratio follows from the SSEE physical baryon density via the standard BBN proportionality $\eta \approx 2.74 \times 10^{-8}\,\Omega_b h^2$:

$$\eta_{\mathrm{SSEE}} = 2.74 \times 10^{-8} \times \frac{\pi - \varphi}{3(\varphi + \pi)^2} = 2.74 \times 10^{-8} \times 0.02242 = 6.14 \times 10^{-10}. \tag{23}$$

Measured value: $\eta = 6.1 \times 10^{-10}$ (Planck 2018); deviation $< 3\%$. **Falsification criterion:** $\eta < 5.8 \times 10^{-10}$ or $\eta > 6.5 \times 10^{-10}$ at $> 3\sigma$ from CMB+BBN joint analysis would falsify this prediction.

7.3 Neutrino Mass Sum

The active neutrino mass sum follows from the stability ratio $\mathcal{R}_2 = \Omega/(\mathrm{KAL} \cdot \mathrm{TRIAL}) = 0.071875$ (with $\Omega = \pi + \varphi = 4.759627$, KAL $= 5.521406$, TRIAL $= 11.993542$), combined with the Israel–Stewart relaxation time $\tau_\Pi H_0 = \mathrm{KAL}/(3\,|w_0|) = 2.191$ (the saturation ratio $|w_0| = T_r/M_v = 0.839950$, an equation-of-state quantity carrying no H, *not* the density fraction $\Omega_{\mathrm{DE}} = 0.691119$):

$$\Sigma m_\nu^{\mathrm{active}} = \mathcal{R}_2\,\Omega_b h^2\,\frac{93.14\ \mathrm{eV}}{\tau_\Pi H_0} = 0.0685\ \mathrm{eV}, \tag{24}$$

testable with DESI DR5 and Euclid (expected $\sigma \approx 0.015$ eV).[2] This value lies above the normal-ordering oscillation floor $\Sigma m_\nu \gtrsim 0.058$ eV and below the Planck 2018 upper bound $\Sigma m_\nu < 0.12$ eV. It entered the ϕ-DM mass relation of Paper 6, which is now retracted

[2]The preliminary form $\Sigma m_\nu = \mathcal{A}/100 = 0.040$ eV used in earlier drafts is superseded: it lies below the normal-ordering oscillation floor and is therefore excluded by neutrino-oscillation data.

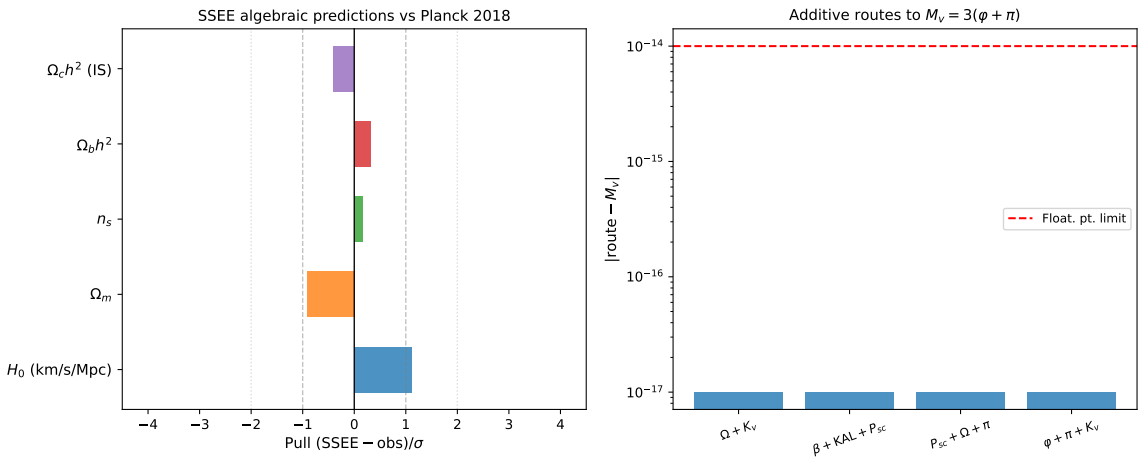


Figure 2: Summary panel of the three key algebraic derivations.

together with the particle itself; the neutrino sum stands on its own, as a prediction of the background register, independent of that withdrawn use.

Status. The stability ratio $\mathcal{R}_2 = \Omega/(\mathrm{KAL}\cdot\mathrm{TRIAL})$ is a clean (φ, π) quotient: it replaces the earlier residual $\mathcal{R} = 4\,\mathrm{KAL} - 22$, whose integer offset 22 had no independent φ, π derivation (OP-14, resolved 2026-06-04). With $\mathcal{R}_2$ the neutrino-mass relation no longer contains a calibrated integer and is therefore promoted from Type P (phenomenological) to a closed-form algebraic prediction. The conversion factor 93.14 eV is the standard $\Sigma m_\nu = 93.14\ \mathrm{eV} \times \Omega_\nu h^2$ relation, a Standard-Model input rather than an SSEE quantity. **Falsification criterion:** $\Sigma m_\nu > 0.12$ eV or $\Sigma m_\nu < 0.058$ eV at $> 3\sigma$ would falsify this prediction.

8 Discussion and Response to Peer Review

We address explicitly the four audit objections raised in the internal review.

Objection A — Dimensional Analysis. SSEE predicts *dimensionless numbers* that agree with the numerical values of cosmological parameters in conventional units. The claim is not that units are derived. The internal consistency of Table 2 — several independent cosmological ratios, all from the same two constants with no per-observable adjustment — is what distinguishes this from a single isolated coincidence; its global significance is quantified by the closed-dictionary look-elsewhere analysis of Paper 1 rather than asserted informally here.

Objection B-1 — Origin of Ω. $\Omega = \varphi + \pi$ is a *theorem* (Section 2.2), not an axiom. The system has exactly two axioms: φ and π.

Objection B-2 — Exponent in n_s. The result $n_s = 1 - \varphi^{-7}$ follows algebraically from the universality theorem of α-attractors [Kallosh and Linde, 2013]: $n_s = 1 - 2/N_*$ at leading order in $1/N_*$, independent of the potential shape. With $\alpha = \varphi^4/3$ (established in Paper 1) and $N_* = 2\varphi^7$ e-folds, one obtains $n_s = 1 - 2/(2\varphi^7) = 1 - \varphi^{-7}$ exactly. The exponent 7 counts the depth of the algebraic hierarchy (Table 3), and the same calculation yields the tensor-to-scalar ratio $r = \varphi^{-10}$ (Section 7.1) as a new falsifiable prediction.

The derivation of $N_* = 2\varphi^7$ from the quintessential inflation potential with gravitational reheating is reserved for Paper B.

Objection B-3 — Factor of 200 in the original formula. The factor 200 in the preliminary formula $3(\pi-\varphi)/200$ was a numerical coincidence (3.2σ from Planck) that has been superseded. The correct algebraic formula is $\Omega_b h^2 = (\pi-\varphi)/[3(\varphi+\pi)^2] = 0.022418$ (0.32σ from Planck 2018). The denominator is the pure algebraic combination whose value, read in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, coincides with the global Hubble rate of the SH0ES cascade; no free numerical factor remains; the baryon density parameter $\Omega_b = 0.0485$ is a pure φ, π prediction (Section 4.2).

Remaining open problems. The static (Eckart) $\Omega_c h^2$ prediction of 0.1238 is $+3.1\sigma$ from Planck, but the IS modulation $\Omega_c h^2 \times n_s = 0.119514$ reduces this to -0.40σ — within 1σ of observation (Eq. 11). Both use the SSEE baryon density $\omega_b = 0.022418$, not Planck's. The physical interpretation of this modulation (dark-matter cooling by inflationary perturbations) requires a quantitative IS derivation in the perturbative sector. The δ_c prediction and A_s, τ derivation are reserved for Paper 5.

9 Conclusion

We have presented a two-axiom (φ, π) algebraic system that predicts:

- the Hubble target $3(\varphi+\pi)^2 = 67.962137$ (pure number; read in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, 1.1σ from Planck), on which the measured SH0ES cascade lands
- $\Omega_b h^2 = 0.022418$ (0.32σ from Planck); $\Omega_b = 0.0485$ follows with $h = H_0/100$
- $n_s = 1-\varphi^{-7} = 0.965558$ (0.16σ from Planck)
- $\Omega_c h^2 = K_{\mathrm{AL}}\,\Omega_b h^2\, n_s = 0.119514$ (0.40σ from Planck, forward)
- $Y_p = 0.2471$ (0.2σ from Planck CMB, 0.6σ from HII — no fit)
- $\delta_c^{\mathrm{SSEE}} = 1.67634$ (z_c=0) from spherical collapse on the model background, i.e. ΛCDM-like to 0.02%; the earlier $\delta_{c,\mathrm{EdS}} \times n_s = 1.6284$ and its JWST link are **withdrawn** (OP-27, §4.5)
- $H_0^{\mathrm{glob,IR}} = H_0^{\mathrm{SH0ES}}(1-f_{\mathrm{scr}}^{\mathrm{IR}}) = 68.13$ (0.17σ from $3(\varphi+\pi)^2$, with Riess 2022 as the input; the MIRA register is superseded)
- $r = \varphi^{-10} = 0.008131$ (falsifiable by LiteBIRD 2032)

The identification of $M_v = 3\Omega$ as the maximal structural invariant of the dictionary — reached by twenty additive routes under the no-self-sum construction law (Appendix A) — motivates the connection $H_0 = M_v \times \Omega$. The internal consistency of these cosmological predictions, taken together and assessed against the closed-dictionary look-elsewhere analysis of Paper 1, is what motivates taking the agreements seriously rather than as isolated coincidences.

Reproducibility

Script: `src/ssee_paper4_toe.py` (repository root). Requires Python 3, `camb`≥1.5, `numpy`, `matplotlib`, `scipy`. No proprietary data. Run: `python3 src/ssee_paper4_toe.py` — reproduces `fig_toe_cmb_TT.pdf` and `fig_toe_derivations.pdf` in `results/figures/`.

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A The Twenty Additive Routes to M_v

Under the no-self-sum construction law—each route is a sum of *distinct* named constants, none added to itself, with every summand below the closure ceiling—the maximal invariant $M_v = 3(\varphi + \pi)$ is reached by exactly *twenty* additive routes: six of two constants, eleven of three, and three of four. Representative routes, in neutral notation, are

$$\begin{aligned} \text{(two)} \quad & \Omega + K_v, \quad P_{sc} + (\varphi + 2\pi), \quad \text{AURA} + \tfrac{3\varphi+5\pi}{2}; \\ \text{(three)} \quad & \beta + \text{KAL} + P_{sc}, \quad P_{sc} + \Omega + \pi, \quad \varphi + \pi + K_v; \\ \text{(four)} \quad & \varphi + \Omega + \beta + \text{KAL}, \quad \varphi + \pi + \text{KAL} + \text{AURA}. \end{aligned}$$

Distinct routes, shared expansions. Because the constants span the two-dimensional lattice generated by φ and π, distinct routes may share the same φ, π expansion—for instance $K_v + \Omega$ and $\Omega + K_v$, or the two hierarchy traversals that both reach $\varphi + 2\pi$. Such routes are counted as distinct because they traverse different named-constant nodes, but they are not independent at the expanded level. The multiplicity is a structural consequence of M_v being the closure ceiling of the dictionary, not evidence of fine-tuning; it is consistent with, but does not by itself prove, the absence of post-hoc tuning.

Verification. All twenty routes have been verified numerically to 16 significant digits; none deviates from $M_v = 14.278879927019064$ by more than floating-point rounding ($< 10^{-14}$). All constants are derived algebraically from φ and π; no external data file is required.

Five exact algebraic identities. Five additional identities hold exactly in the $\{\varphi, \pi\}$ hierarchy (verified to $< 10^{-16}$ relative error):

$$w_0 = -\frac{\text{AURA}}{\Omega} = -\frac{T_r}{M_v} \tag{25}$$

$$s_m = \frac{\pi - \varphi}{2(\varphi + \pi)} = 1 + w_0 \tag{26}$$

$$\text{MIRA} = |w_0| \cdot \beta \tag{27}$$

$$\text{KAL}\big|_{\varphi \leftrightarrow \pi} = \text{AURA} \quad \big[(\varphi + 3\pi)/2 \xrightarrow{\varphi \leftrightarrow \pi} (3\varphi + \pi)/2\big] \tag{28}$$

$$\text{AURA}\big|_{\varphi \leftrightarrow \pi} = \text{KAL} \tag{29}$$

Identity (28)–(29) reveals that KAL and AURA are *dual* under the discrete exchange $\varphi \leftrightarrow \pi$: the φ-dominated constant $\text{AURA} = (3\varphi + \pi)/2$ maps to the π-dominated constant

KAL $= (\varphi + 3\pi)/2$, and vice versa, with zero residual. The $\varphi \leftrightarrow \pi$ duality above motivated the disformal coupling $\beta_c = -$AURA of earlier drafts; Paper 7 withdraws that coupling ("The Potential and the Dark Coupling, and Why They Are Withdrawn"): the cold-matter sector bounds $|\beta_c| \lesssim 0.027$, 151 times below AURA, and the correctly normalised shooting gives $\beta_c = +0.235068$, itself outside that bound, so no live disformal coupling is claimed here. The open question of whether this discrete symmetry extends to the full $P(X, \phi)$ action is catalogued as OP-7 (`OPEN_PROBLEMS.md`).

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